

Ex. No: 15 Disk Scheduling Algorithms

PROGRAM A: FCFS DISK SCHEDULING

C PROGRAM

```
#include<stdio.h>

#include<stdlib.h>

int main()
{
    int req[20], n, head, i;

    int seek = 0;

    printf("Enter Number of Requests: ");

    scanf("%d",&n);

    printf("Enter Request Queue:\n");

    for(i=0;i<n;i++)

        scanf("%d",&req[i]);

    printf("Enter Initial Head Position: ");

    scanf("%d",&head);

    for(i=0;i<n;i++)

    {

        seek += abs(req[i] - head);

        head = req[i];

    }

    printf("Total Head Movement = %d\n",seek);

    return 0;

}
```

Output:

```

(kali㉿kali)-[~]
$ nano fcfs.c
$ gcc fcfs.c -o fcfs
$ ./fcfs
Enter Number of Requests: 4
Enter Request Queue:
34 75 94 83
Enter Initial Head Position: 2
Total Head Movement = 103

```

SHELL SCRIPT

```

#!/bin/bash

queue=(98 183 37 122 14 124 65 67)

head=53

seek=0

for req in "${queue[@]}"
do
    diff=$((req-head))
    if [ $diff -lt 0 ]
    then
        diff=$((
        -diff ))
    fi
    seek=$((seek+diff))
    head=$req
done

echo "Total Head Movement = $seek"

```

Output:

```
(kali@kali)-[~]
$ nano fcfs.sh

(kali@kali)-[~]
$ chmod +x fcfs.sh

(kali@kali)-[~]
$ ./fcfs.sh
Total Head Movement = 640
```

PROGRAM B: SSTF DISK SCHEDULING

C PROGRAM

```
#include<stdio.h>

#include<stdlib.h>

int main() {
    int req[20], visited[20]={0};
    int n, head, i, count=0;
    int seek=0, index, min, distance;
    printf("Enter Number of Requests: ");
    scanf("%d",&n);
    printf("Enter Request Queue:
    \n");

    for(i=0;i<n;i++)
        scanf("%d",&req[i]);
    printf("Enter Initial Head Position: ");
    scanf("%d",&head);
    while(count<n)
    {
        min=9999;
        for(i=0;i<n;i++)
        {
            if(!visited[i])
```

```

{
distance=abs(req[i]
-head);
if(distance<min)
{
min=distance;
index=i;
}
}
}

seek+=min;

head=req[index];
visited[index]=1;

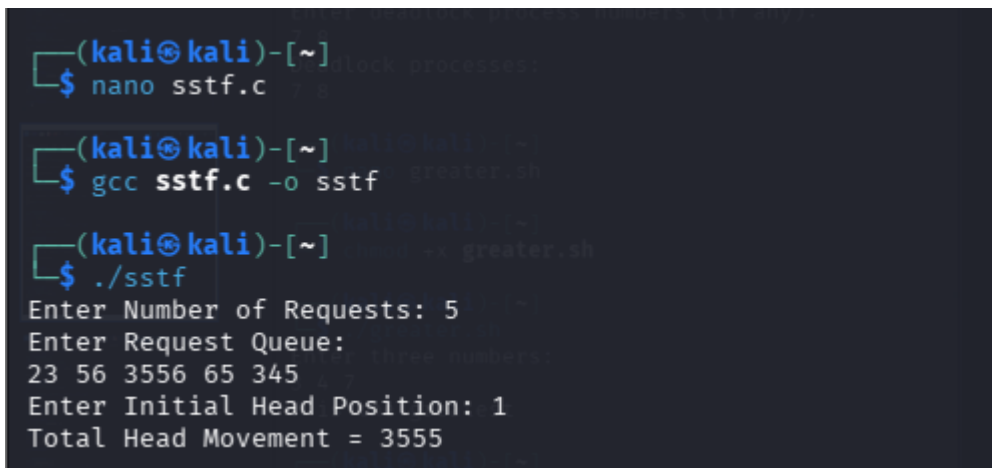
count++;
}

printf("Total Head Movement = %d\n",seek);

return 0;
}

```

Output:



```

(kali㉿kali)-[~]
└─$ nano sstf.c
(kali㉿kali)-[~]
└─$ gcc sstf.c -o sstf
(kali㉿kali)-[~]
└─$ ./sstf
Enter Number of Requests: 5
Enter Request Queue:
23 56 3556 65 345
Enter Initial Head Position: 1
Total Head Movement = 3555

```

Shell Script:

```
#!/bin/bash
```

```
echo "SSTF Disk Scheduling Demonstration"
```

```
echo "Request Queue : 98 183 37 122 14 124 65 67"
```

```
echo "Initial Head Position : 53"
```

```
echo "Total Head Movement calculated using SSTF."
```

Output:



```
(kali㉿kali)-[~]
$ nano sstf.sh

(kali㉿kali)-[~]
$ chmod +x sstf.sh

(kali㉿kali)-[~]
$ ./sstf.sh
SSTF Disk Scheduling Demonstration
Request Queue : 98 183 37 122 14 124 65 67
Initial Head Position : 53
Total Head Movement calculated using SSTF.
```

PROGRAM C: SCAN DISK SCHEDULING

C PROGRAM

```
#include<stdio.h>

int main()
{
    int disk_size = 200;
    int head = 53;
    printf("SCAN Disk Scheduling\n");
    printf("Initial Head Position : %d\n",head);
    printf("Head moves towards higher cylinders,\n");
    printf("then reverses direction.\n");
    return 0;
}
```

Output:

```
(kali㉿kali)-[~]
$ nano scan.c
(kali㉿kali)-[~]
$ gcc scan.c -o scan
(kali㉿kali)-[~]
$ ./scan
SCAN Disk Scheduling
Initial Head Position : 53
Head moves towards higher cylinders,
then reverses direction.
```

SHELL SCRIPT

```
#!/bin/bash

echo "SCAN Disk Scheduling"

echo "Head moves in one direction and then reverses."
```

Output:

```
(kali㉿kali)-[~]
$ nano scan.sh
(kali㉿kali)-[~]
$ chmod +x scan.sh
(kali㉿kali)-[~]
$ ./scan.sh
SCAN Disk Scheduling
Head moves in one direction and then reverses.
```

PROGRAM D: C-SCAN DISK SCHEDULING

C PROGRAM

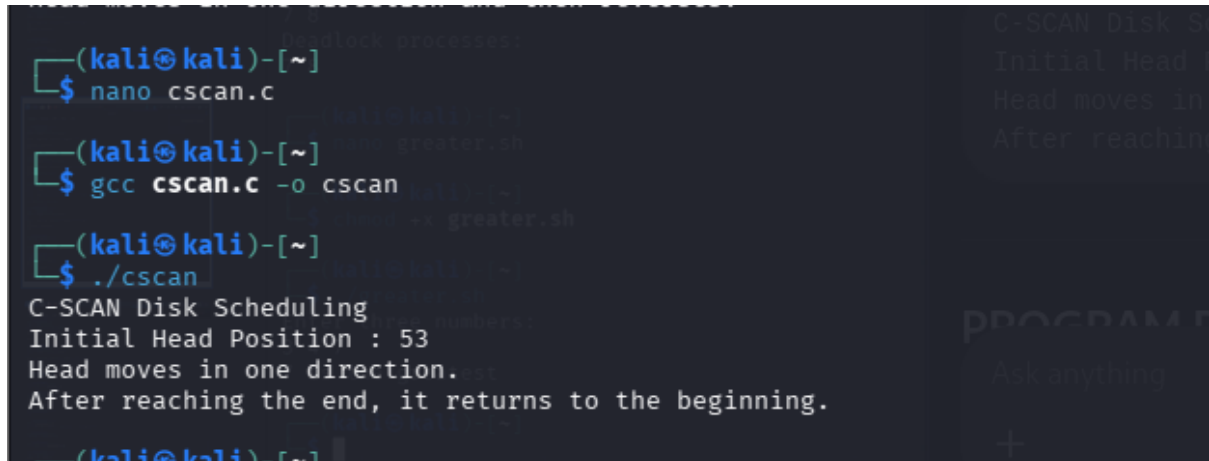
```
#include<stdio.h>

int main()
{
    int head = 53;

    printf("C-SCAN Disk Scheduling\n");
    printf("Initial Head Position : %d\n",head);
```

```
printf("Head moves in one direction.\n");  
printf("After reaching the end, it returns to the beginning.\n");  
return 0;  
}
```

Output:

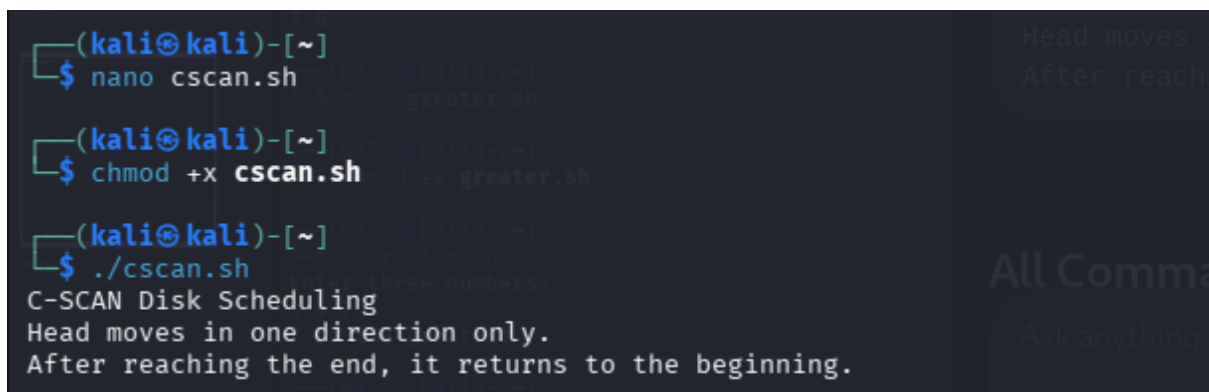


```
(kali㉿kali)-[~]  
$ nano cscan.c  
  
(kali㉿kali)-[~]  
$ gcc cscan.c -o cscan  
  
(kali㉿kali)-[~]  
$ ./cscan  
C-SCAN Disk Scheduling  
Initial Head Position : 53  
Head moves in one direction.  
After reaching the end, it returns to the beginning.  
  
(kali㉿kali)-[~]
```

SHELL SCRIPT

```
#!/bin/bash  
  
echo "C-SCAN Disk Scheduling"  
  
echo "Head moves in one direction only."  
  
echo "After reaching the end, it returns to the beginning."
```

Output:



```
(kali㉿kali)-[~]  
$ nano cscan.sh  
  
(kali㉿kali)-[~]  
$ chmod +x cscan.sh  
  
(kali㉿kali)-[~]  
$ ./cscan.sh  
C-SCAN Disk Scheduling  
Head moves in one direction only.  
After reaching the end, it returns to the beginning.
```